RAIHAN AHMAD

Software Engineer | Full-Stack Developer

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SUMMARY

Computer Science undergraduate with production experience in Python, FastAPI, React/TypeScript and PostgreSQL, with end-to-end ownership of features in a live multi-tenant platform. Strong in data structures, algorithms and object-oriented design, with hands-on experience building LLM tool-calling and agent-native systems.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL
Backend & LLM: FastAPI, SQLAlchemy 2.0, Alembic, Pydantic, REST APIs, LLM Tool Calling
Frontend: React.js, Material UI, Tailwind CSS
Databases: PostgreSQL, MySQL, MongoDB, SQLite
Auth, DevOps & Tools: Keycloak, OAuth2, Docker, Git, GitHub Actions (CI), Linux, Playwright, pytest

EXPERIENCE

Software Developer Intern | SkyGrid LLC, Dubai, UAE (Remote) Aug 2026 - Present

- Built Benefits Configuration end to end: 50 FastAPI endpoints, 17 PostgreSQL tables and 6 Alembic migrations covering components, salary structures, benefit plans and employee assignments.
- Implemented reusable eligibility and calculation engines (11 eligibility fields, 22 field/operator combinations, 5 formula node types) and 4 React/TypeScript configuration screens.

Software Developer Intern | Nucelux Private Limited, New Delhi, India (Remote) Mar 2026 - Jul 2026

SkyGrid - Enterprise HRMS & Payroll Platform (Nucelux partner product)

- Delivered 12 modules end to end (PostgreSQL schema, Alembic migrations, business logic, REST APIs, React/TypeScript UI; ∼100 endpoints) in a live multi-tenant platform of ∼96,500 backend lines and 620 endpoints built by 8 engineers.
- Built payroll-critical features: a 12-endpoint Final Settlement (end-of-service) engine on the payroll batch calculation, a validated Variable Earnings bulk-upload pipeline with duplicate detection, and Air Ticket Processing with journal vouchers.
- Re-architected the monthly Roster (V2) around a pure domain layer; introduced the project's first frontend testing framework, authoring 7 of 9 test files.
- Audited department-scoped RBAC: found it enforced in only 1 of 9 modules, with one scope type failing open instead of closed; reproduced the issue against real data and wrote the remediation spec.
- Repaired the shared Alembic migration graph (∼200 branches, 8 developers), fixed Docker and CI build failures, and authored ∼4,100 lines of engineering documentation.

Full Stack Developer Intern | Zidio Development, Bengaluru, India (Remote) Mar 2025 - Apr 2025

- Led front-end development of FinTrak, a finance-tracking app, building a reusable React.js and Material UI component library.
- Built Spring Boot REST APIs over MySQL with JWT authentication and Chart.js analytics dashboards.

PROJECTS

FoodHub - Agent-Native Food Ordering Platform | GitHub Jul 2026 - Sep 2026

- Built a FastAPI service exposing 5 agent capabilities through a discoverable manifest for Gemini tool calling, with a server-side policy engine that allows, holds for approval or denies each action.
- Implemented per-order and rolling daily spending limits, record-level authorization, audit logging and a 680-test suite; in 24 benchmark trials, completed agent-API runs used ∼20x fewer tokens than Playwright browser-agent runs.

RouteLab - Delivery Routing & Optimization | droptimizer.netlify.app Mar 2026 - Apr 2026

- Built a delivery route optimization platform with Python/FastAPI, React/TypeScript and OSRM, exposing 23 API handlers that assign deliveries to vehicles under capacity, time-window and shift constraints.
- Implemented 5 selectable routing solvers (cheapest insertion, local search, ALNS, exact reference, OR-Tools) supporting up to 100 customers and 25 vehicles, with durable background job execution.

Student Stress Detector - Fuzzy Inference System | stressdetect.netlify.app Nov 2025 - Dec 2025

- Built a full-stack student stress assessment platform with FastAPI, React/TypeScript and PostgreSQL, powered by a deterministic 3-component, 27-rule fuzzy inference engine with explainable, contributor-based results.
- Validated with 248 automated tests across 1,375 supported input combinations, achieving 2.42 ms p95 inference latency, with longitudinal history and isolated what-if analysis.

EDUCATION

B.Tech in Computer Science and Engineering | Birla Institute of Technology, Mesra | CGPA: 8.51/10 2023 - 2027

Holy Cross School, Hazaribagh (CBSE) | Class XII: 91% | Class X: 95.5% 2022, 2020

LEADERSHIP & ACTIVITIES

- Google Student Ambassador (Aug 2025 - Dec 2025): delivered hands-on AI workshops to 300+ students.
- General Secretary, Aakshar Literary Society (2025): directed 15+ campus events, expanding outreach.
- School Captain, Holy Cross School (2020 - 2022): led the student council of a 1,000+ student school.